Literature-implied partial answer: the Laakso-to-tree question for dual Banach spaces

Source: M. I. Ostrovskii, arXiv:1102.5082
Supporting paper: M. I. Ostrovskii, arXiv:1302.5968

Status. literature_implied_answer (partial subcase). The implication below settles the source's question for dual Banach spaces, but not for arbitrary Banach spaces.

Source question. In arXiv:1102.5082, Theorem `T:DiamTree` proves that if the infinite diamond D_ω embeds bilipschitzly into a Banach space X, then X contains a bounded δ-tree for some $\delta > 0$. Later, in the proof of Theorem `T:LaakTree`, the paper proves that bilipschitz embeddability of the Laakso space L_ω into X implies that X contains a bounded δ-semitree and hence fails the Radon–Nikodym property. At that point the paper states that it is not known whether bilipschitz embeddability of L_ω into X implies the existence of a bounded δ-tree in X.

Supporting theorem. The later paper arXiv:1302.5968 proves in Theorem 1.10 that a dual Banach space does not have the Radon–Nikodym property if and only if it admits a bilipschitz embedding of D_ω.

Theorem. *Let X be a dual Banach space. If L_ω embeds bilipschitzly into X, then X contains a bounded δ-tree for some $\delta > 0$.*

Proof. Assume that L_ω embeds bilipschitzly into the dual Banach space X. By arXiv:1102.5082, Theorem `T:LaakTree`, the space X does not have the Radon–Nikodym property. Since X is a dual Banach space, arXiv:1302.5968, Theorem 1.10, implies that D_ω embeds bilipschitzly into X. Applying arXiv:1102.5082, Theorem `T:DiamTree`, to this embedding of D_ω, we obtain a bounded δ-tree in X for some $\delta > 0$. □

Scope and provenance. This is not a full answer to the source question, because the dual-space hypothesis is essential to the supporting theorem used here. The relation is an agent-identified implication from a later theorem; arXiv:1302.5968 does not explicitly state that it answers the sentence in arXiv:1102.5082.

References

[1] M. I. Ostrovskii, *On metric characterizations of some classes of Banach spaces*, arXiv:1102.5082.

[2] M. I. Ostrovskii, *On metric characterizations of the Radon-Nikodym and related properties of Banach spaces*, arXiv:1302.5968.